

APPLIED BEHAVIOR ANALYSIS

CHARACTERISTICS OVERVIEW CHART

Verbal Skills	Grade Levels	Cognitive Level	Areas Addressed
☒ Nonverbal	☒ PK	☒ Classic	☒ (Pre)Academic/Cognitive/Academic
☒ Mixed	☒ Elementary	☒ High Functioning	☒ Adaptive Behavior/ Daily Living
☒ Verbal	☒ Middle/High		☒ Behavior ☒ Communication/Speech ☒ Social/Emotional

BRIEF INTRODUCTION

Applied behavior analysis (ABA) has its roots in the philosophy of modern behaviorism pioneered by Skinner, who laid the foundation in the early 1900s. As such, it is the practical application of behavioral laws, laws of reinforcement, formulated by Skinner to solve behavior problems. ABA is defined as the process of systematically applying interventions based upon the principles of behavior theory to improve socially significant behaviors, including reading, academics, social skills, communication, and adaptive living skills, to a meaningful degree, and to demonstrate that the interventions employed are responsible for the improvement in behavior (Baer, Wolf, & Risley, 1968; Sulzer-Azaroff & Mayer, 1991).

DESCRIPTION

According to Baer et al. (1968), ABA is “the process of applying sometimes tentative principles of behavior to the improvement of specific behaviors, and simultaneously evaluating whether or not any changes noted are indeed attributed to the process of application” (p. 91). ABA-based interventions have been popular with children with autism since the 1980s and can be linked to the work of Ivar Lovaas (1987).

Numerous studies have demonstrated and documented the effectiveness of ABA-based intervention with individuals with autism. To date, ABA-based interventions have been used to support individuals with autism in at least six ways (Shaping Behavior, 2008):

1. To increase behavior;

2. To teach a new skill;
3. To maintain behaviors;
4. To generalize or to transfer behavior from one situation or response to another;
5. To restrict or narrow conditions under which interfering behaviors occur; and
6. To reduce interfering behavior.

According to Baer et al. (1968) and Heward, Heron, Neef, Peterson, Sainato, Cartledge, et al. (2005), ABA has the following characteristics:

Applied. ABA focuses on areas that are of social significance. Behavior scientists not only take into consideration long-term behavior change, but also look at how behavior changes affect the individual and the people around.

Behavioral. The behavior must be objectively measured and must change.

Analytic. The behavior scientist must demonstrate control over the behavior with intervention while maintaining ethical standards.

Systematic. ABA interventions and procedures must be very detailed so that other researchers are able to replicate the application with the same results.

Generalizeable. The results of an ABA-based intervention must last over time, across settings, and spread to other behaviors not directly addressed.

Data-based. ABA uses direct and frequent measurements to enable analysts to detect their success or failures so that appropriate changes can be made.

Many programs use some or all of the components of ABA, and sometimes are referred to ABA programs. However, calling a program ABA does not indicate which specific interventions are used. Indeed, there is no single intervention called ABA.

Programs that are commonly referred to as having ABA-based interventions include (a) Young Autism Project or Lovaas Model (Lovaas, 1987), (b) Princeton Child Development Institute (Fenske, Zalenski, Krantz, & McClannahan, 1985), (c) Douglass Developmental Disabilities Center at Rutgers University (Handleman, Harris, Gordon, Kristoff, & Fuentest, 1991), (d) the River Street Autism Program (Dyer, Martino, & Parvenski, 2006), (e) Therapeutic Pathways (Howard, Sparkman, Howard, Green, & Stanislaw, 2005), (f) Walden Program (McGee, Morrier, & Daly,

1999), (g) the Institute for Child Development at State University of New York (Romanczyk, Lockshin, Gillis, & Matey, 2006), (h) Autism Partnerships (Leaf, McEachin, & Harsh, 1999), (i) May Institute (Luce & Christian, 1989), (j) Pivotal Response Treatment (Koegel, Koegel, Shoshan, & McNerney, 1999), (k) Alpine Learning Group (Meyer, Yaylor, Cerion, Fisher, Moran, & Richard, 2006), (l) Pyramid Approach (Bondy & Battaglini, 2006), (m) Verbal Behavior (Partington, 2005), (n) Eden Model (Holmes, 1998), (o) STAR Program (Arick, Krug, Loos, & Falco, 2005), (p) Learning Experiences and Alternative Program for Preschoolers and Their Parents (LEAP; Strain & Hoyson, 2000), (q) Project Data (Schwartz, Sandall, McBride, & Boulware, 2004), and (r) Gentle Teaching (McGee, Menolascino, Hobbs, & Menousek, 1987).

SUMMARY

ABA is a framework for the practice of a science that systematically applies interventions based upon the principles of behavior theory to improve socially significant behaviors. It emphasizes explaining behavior in terms of external events that can be manipulated. Techniques and principles associated with ABA include discrete trial training, errorless learning, incidental teaching, prompting, fading, shaping, etc. The effectiveness of individual ABA-based interventions with individuals with autism is well documented.

RESEARCH TABLE

No table is presented here as ABA is not a single intervention. Please refer to individual interventions listed under the description.

REFERENCES

- Arick, J., Krug, D., Loos, L., & Falco, R. (2005). School-based programs. In F. Volkmar, R. Paul, A. Klin, & D. Cohen (Eds.), *Handbook of autism and pervasive developmental disorders*. Hoboken, NJ: John Wiley & Sons.
- Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behavior analysis. *Journal of Applied Behavior Analysis*, 1, 91-97.
- Bernard-Opitz, V., Ing, S., & Kong, T. Y. (2004). Comparison of behavioural and natural play interventions for young children with autism. *Autism*, 8, 319-333.

- Bondy, A., & Battaglini, K. (2006). Application of the pyramid approach to education model in a public school setting. In J. Handleman & S. Harris (Eds.), *School age education programs for children with autism*. Austin, TX: Pro-Ed.
- Dyer, K., Martino, G., & Parvenski, T. (2006). The River Street Autism Program: A case study of a regional service center behavioral intervention program. *Behavior Modification*, 30, 925-943.
- Fenske, E., Zalsenski, S., Krantz, P., & McClannahan, L. (1985). Age at intervention and treatment outcome for autistic children in a comprehensive intervention program. *Analysis and Intervention in Developmental Disabilities*, 5, 49-58.
- Handleman, J., Harris, S., Gordon, R., Kristoff, B., & Fuentes, F. (1991). Changes in cognitive and language functioning of preschool children with autism. *Journal of Autism and Developmental Disorders*, 21, 281-290.
- Heward, W. L., Heron, T. E., Neef, N. A., Peterson, S. M., Sainato, D. M., Cartledge, G., et al. (Eds.). (2005). *Focus on behavior analysis in education: Achievements, challenges, and opportunities*. Upper Saddle River, NJ: Prentice Hall/Merrill.
- Holmes, D. (1998). *Autism through the lifespan: The Eden Model*. Bethesda, MD: Woodbine House.
- Howard, J., Sparkman, C., Howard, C., Green, G., & Stanislaw, H. (2005). A comparison of intensive behavior analytic and eclectic treatments for young children with autism. *Research in Developmental Disabilities*, 26, 359-383.
- Koegel, L. K., Koegel, R. L., Shoshan, Y., & McNerney, E. (1999). Pivotal response intervention II: Preliminary long-term outcomes data. *Journal of the Association for Persons with Severe Handicaps*, 24, 186-198.
- Leaf, R., McEachin, J., & Harsh, J. (1999). *A work in progress: Behavior management strategies and a curriculum for intensive behavioral treatment of autism*. New York: DRL Books.
- Lovaas, O. I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. *Journal of Consulting and Clinical Psychology*, 55, 3-9.
- Luce, S., & Christian, W. (1989). State-of-the-art programming in Massachusetts: A brief description of the May Institute. In Christopher Gillberg (Ed.), *Diagnosis and treatment of autism*. New York: Plenum.
- McGee, J. J., Menolascino, F. J., Hobbs, D. C., & Menousek, P. E. (1987). *Gentle teaching: A non-aversive approach to helping persons with mental retardation*. New York: Human Sciences Press.

- McGee, G., Morrier, M., & Daly, T. (1999). An incidental teaching approach to early intervention for toddlers with autism. *The Journal of the Association for Persons with Severe Handicaps*, 24, 133-146.
- Meyer, L., Taylor, B., Cerion, K., Fisher, J., Moran, L., & Richard, E. (2006). Alpine Learning Group. In J. Handleman & S. Harris (Eds.), *School age education programs for children with autism*. Austin, TX: Pro-Ed.
- Partington, J. (2005). *The updated ABLLS: The Assessment of Basic Language and Learning Skills*. Pleasant Hill, CA: Behavior Analysts, Inc.
- Romanczyk, R. G., Lockshin, S., Gillis, J. M., & Matey, L. (2006). Institute for Child Development – preschool program. In S. Harris & J. Handleman (Eds.), *Preschool programs for children with autism* (3rd ed.). Austin, TX: Pro-Ed.
- Schwartz, I. S., Sandall, S. R., McBride, B. J., & Boulware, G. L. (2004). Project DATA (Developmentally Appropriate Treatment for Autism): An inclusive school-based approach for young children with autism. *Topics in Early Childhood Education*, 24(3), 156-168.
- Shaping Behavior. (n.d.). Retrieved October 25, 2008, from Shaping Behavior Web site: <http://www.shapingbehavior.com>
- Strain, P., & Hoyson, M. (2000). The need for longitudinal intensive social skill intervention: LEAP follow-up outcomes for children with autism. *Topics in Early Childhood Special Education*, 20, 116-123.
- Sulzer-Azaroff, B., & Mayer, R. (1991). *Behavior analysis of last changes*. Fort Worth, TX: Holt, Reinhart & Winston, Inc

RESOURCES AND MATERIALS

- Association for Behavior Analysis International: www.abainternational.org/ba.asp
This link takes the user directly to the behavior analysis portion of the larger website.
- Autism Intervention Information: <http://knospe-aba.com/cms/en/home.html>
The goal of this website is to be a resource for parents; it includes links to other resources.
- B. F. Skinner Foundation: www.bfskinner.org
Though non-functional at the time of this publication, this site includes a downloadable program for instructors to use in the classroom.
- Behavior Analyst Certification Board (BACB): <http://www.bacb.com/>
Part of the website for the BACB, this link directs the user to an overview of ABA. Other links on this website include a list of board certified analysts and helpful consumer information.

- Fovel, J. Tyler (2002). *The ABA program companion: Organizing quality programs for children with autism and PDD*. New York: DLR Books, Inc.
This excellent new manual and CD is geared towards helping the reader integrate important theories and concepts from ABA into powerful, practical and comprehensive educational programming, from assessment through program methodology and evaluation of results. Topic summaries, key concepts, question lists and references are found throughout the manual, along with reproducible data forms in order to organize and expedite program development.
- Harris, S. and Weiss, M.J. (2007). *Right from the start: Behavioral intervention for young children with autism*. 2nd Ed. Bethesda, MD: Woodbine House.
Parents of young children--ages 18 months to 5 years--recently diagnosed with autism, face the critical decision of which educational treatment program to choose for their child. To help guide them through this complex decision-making process, parents can turn to the revised and updated edition of *Right from the Start*. This second edition covers programs that use Early Intensive Behavioral Intervention, a highly effective treatment based on Applied Behavior Analysis (ABA).
- Lovaas, O. I. (1977). *The autistic child: Language development through behavior modification*. New York: Irvington Press.
- Lovaas, O. L. (2003). *Teaching individuals with developmental delays: Basic intervention techniques*. Austin, TX: Pro-Ed.
- Miller, L. K. (2006). *Principles of every day behavior analysis* (4th ed.). Belmont, CA: Thomson Wadsworth.
These three books discuss the various applied behavior analysis strategies.
- Mueller, M. M., Palkovic, C. H., & Maynard, C. S. (2007). Errorless learning: Review and practical application for teaching children with pervasive developmental disorders. *Psychology in the Schools*, 44(7), 691-700.
This journal article reviews the literature on errorless learning and provides examples.
- Polyxo.com: Teaching Children with Autism: <http://www.polyxo.com/>
This teacher resource offers foundational concepts for developing an ABA program.
- Pratt, C. & Dubie, M. (2008). "Observing Behavior Using A-B-C Data." IRCA Articles. <http://www.iidc.indiana.edu/irca/behavior/abcData.html>

GENERAL RESOURCES

- Autism Internet Modules (AIM) www.autisminternetmodules.org
The Autism Internet Modules were developed with one aim in mind: to make comprehensive, up-to-date, and usable information on autism accessible and applicable to educators, other professionals, and families who support individuals with autism spectrum disorders (ASD). Written by experts from across the U.S., all online modules are free, and are

designed to promote understanding of, respect for, and equality of persons with ASD.

Current modules are:

- Assessment for Identification
 - Home Base
 - Peer-Mediated Instruction and Intervention (PMII)
 - Picture Exchange Communication System (PECS)
 - Pivotal Response Training (PRT)
 - Preparing Individuals for Employment
 - Reinforcement
 - Restricted Patterns of Behavior, Interests, and Activities
 - Self-Management
 - Social Supports for Transition-Aged Individuals
 - Structured Teaching
 - Structured Work Systems and Activity Organization
 - Supporting Successful Completion of Homework
 - The Incredible 5-Point Scale
 - Time Delay
 - Transitioning Between Activities
 - Visual Supports
- Interactive Collaborative Autism Network (iCAN) <http://www.autismnetwork.org>
iCAN offers free online instructional modules on autism spectrum disorder (ASD). Modules have been developed in these areas:
 - Characteristics
 - Assessment
 - Academic Interventions
 - Behavioral Interventions
 - Communication Interventions
 - Environmental Interventions
 - Social Interventions
- Indiana Resource Center for Autism (IRCA) <http://www.iidc.indiana.edu/irca/fmain1.html>
The Indiana Resource Center for Autism staff's efforts are focused on providing communities, organizations, agencies, and families with the knowledge and skills to support children and adults in typical early intervention, school, community, work, and home settings.
 - IRCA Articles <http://www.iidc.indiana.edu/irca/ftrainpapers.html>
 - IRCA Modules <http://www.iidc.indiana.edu/irca/fmodules.html>
- Texas Statewide Leadership for Autism www.txautism.net
The Texas Statewide Leadership for Autism in conjunction with the network of Texas Education Service center with a grant from the Texas Education Agency has developed a series of free online courses in autism. Please check the training page, www.txautism.net/training.html, for update lists of courses, course numbers and registration information. Current courses include the following:

- Autism 101: Top Ten Pieces to the Puzzle
- Autismo 101: Las 10 piezas principales del rompecabezas
- Asperger Syndrome 101 Online
- Asperger Syndrome 101 Online
- Navigating the Social Maze: Supports & Interventions for Individuals with Autism Spectrum Disorders
- Communication: The Power of Communication for Individuals with Autism Spectrum Disorders
- Communication: The Power of Communication for Individuals with Autism Spectrum Disorders



Applied Behavior Analysis in Special Education

Misconceptions and Guidelines for Use

Cary E. Trump, Robert C. Pennington, Jason C. Travers,
Joel E. Ringdahl, Erinn E. Whiteside, and Kevin M. Ayres

After returning from a conference, Chantelle, an undergraduate student majoring in special education, finds that she has many questions related to the use of applied behavior analysis (ABA) within teaching. During the conference, she attended a few exciting sessions describing the positive effects of behavior-analytic practices in special education. However, she also attended sessions in which presenters did not speak favorably about ABA, and some even regarded it as a cruel practice for students with disabilities. She sought guidance from her advisor, Dr. Thatcher, to answer her questions and help her decide whether or not she should incorporate ABA into her future practice.

ABA involves the application of the science of human behavior to address important social issues (Cooper, Heron, & Heward, 2007). It encompasses a range of practices and procedures that have been well established in the research literature as effective and in many cases have been deemed as “evidence based,” such as discrete trial teaching, reinforcement, the Picture Exchange Communication System, and task analyses (Wong et al., 2015). Researchers and practitioners in recent years have provided many examples of the utility of ABA for addressing educational and behavioral needs of students receiving special education services. For example, positive reinforcement has been widely used in programs designed to teach students in general and special education a variety of skills, both adaptive and academic (Alberto & Troutman, 2013). Research teams have applied positive reinforcement to teach communication (Cagliani, Ayres, Whiteside, & Ringdahl, 2017), reading (Carroll, Joachim, St. Peter, & Robinson, 2015), employment skills (Collins, Terrell, & Test, 2017), and a vast range of other instructional targets. Group interventions, based on the principles of ABA, also are implemented in general education settings to support all students, including those with and without disabilities (e.g., Kamps et al., 2011). In fact, many highly effective practices in special education are

derived from ABA (e.g., explicit instruction, response prompting, positive feedback; Figure 1).

Throughout the conference, Chantelle noticed almost all of the ABA-related presentations focused on students with autism spectrum disorder (ASD). She wondered if this approach was relevant only for students with ASD. During their meeting, Dr. Thatcher explained the connection between ABA and ASD.

ABA and ASD

Despite the widespread applicability of ABA for the instruction and support of students with a range of abilities, it has become synonymous with interventions designed for students with ASD. The strong association between ABA and ASD is mostly likely due to a confluence of factors. First, the work of an ABA pioneer, Ivar Lovaas, demonstrated significant social-communication gains for individuals with ASD following early, intensive behavioral intervention (Lovaas, 1987). As a

result, many intervention programs for children with ASD adopted behavioral strategies as their primary approach for teaching communication (e.g., picture exchange) and academic skills (e.g., constant time delay). Second, within the past decade, an increasing number of health insurance companies began to provide coverage for the provision of therapeutic services to individuals with ASD (Baller et al., 2016). Although instrumental in helping individuals access evidence-based therapeutic services, providing insurance coverage for ABA services for individuals with ASD created incentives for practitioners to specifically market their services toward that population. Finally, strong parent advocacy groups have played a role in promoting the use of ABA for students with ASD (Guercio & Murray, 2014). Although there is a clear link between delivery of effective instruction and support of students with ASD and ABA, ABA’s applications go beyond this population (e.g., Roane, Ringdahl, & Falcomata, 2015).

Figure 1. Examples of ABA applications beyond autism spectrum disorder.

ABA Utility in Schools Beyond ASD
Problem behavior reduction: • Aggression • Bullying • Disruption • Off-task behaviors • Property destruction Skill acquisition: • Appropriate communication strategies • Improved student performance in math, reading, and writing • Safety skills • Social skills Teaching strategies: • Behavior skills training • Descriptive praise statements • Direct instruction • Opportunities to respond • Self-monitoring • Task analysis • Video modeling

Dr. Thatcher helped Chantelle realize that ABA is not a specific set of interventions for students with ASD; rather, ABA is a scientific approach to teaching and learning that could be useful for teaching academic, social, and other skills to a wide range of learners.

Chantelle still had questions, though. She had heard ABA proponents were more interested in collecting and graphing data than a student's welfare; however, some parents did not believe data sets and graphs could adequately represent their child. Wasn't a child more complex?

Behavior Analysis and Data Collection

Behavior analysis is particularly well known for its heavy reliance on data collection and evaluation. The reliance on data is justified by an ethical obligation to ensure students with disabilities are provided services that include access to effective interventions and therapeutic environments (Council for Exceptional Children, 2015; Van Houten, Axelrod, Bailey, & Favell, 1988). Specifically, the objective measurement of student progress is crucial for ensuring the provision of effective instruction because recollection and subjective impressions (i.e., reflection) are unreliable (Fixsen, Phillips, & Wolf, 1972). A behavior analyst will likely coach a teacher to track, in real time, the number of minutes a student is out of seat because the teacher's memory might be influenced by a range of classroom variables (e.g., behavior of other students, increased attention to the target student, focus on the lesson). The data are then graphed, visually analyzed, and used as evidence for determining intervention effectiveness—whether an intervention is working. Using data to conduct objective evaluations of intervention effects on both targeted (e.g., in-seat) and nontargeted (e.g., talking-out) behavior is consistent with the ethical underpinnings of behavior analysis and special education. Thus, in contrast to misconceptions related to impersonal

use of data, data collection is an essential aspect of individualizing intervention and support as well as minimizing subjectivity.

Although there is a clear link between delivery of effective instruction and support of students with ASD and ABA, ABA's applications go beyond this population.

Chantelle learned that the emphasis on data collection in the behavior-analytic community was no different from current teaching practices like progress monitoring that are used to accurately measure and track student performance. She also realized the importance of data in making decisions about whether to intervene, to continue a current intervention, or to make adjustments. Most important, Chantelle understood that the data were not a representation of the whole student per se, but rather data collection was a tool to support intervention decisions and advocate for the student receiving services. Although Chantelle was feeling a bit better about ABA now that she had clarified some misperceptions, she was still concerned about the use of punishment in ABA. She had read articles from the 1960s and 1970s where aversive punishment was used and wondered if and how punishment is used today.

Behavior Analysis and the Use of Punishment

Punishment involves the delivery of aversive stimuli (i.e., an undesired item or activity) or the removal of preferred stimuli following a behavior that decreases the likelihood that behavior will occur again in the future (Cooper et al., 2007). Although there are many behaviors requiring the use of behavior reduction procedures (e.g., aggression, self-injurious behaviors, bullying), ABA has a history of intervention procedures that are considered crude and unethical by today's standards—including the use of procedures such as restraint and electric shock (Bailey &

Burch, 2011). Similar to other professions, there are some bad actors that misuse techniques in ways that are harmful to others. Although

disconcerting, the examples involving inappropriate use of behavior-analytic techniques suggest professionals are not immune to fallibilities associated with human workers. Recognizing the potential for human fallibility does not suggest that practitioners should not be held accountable for their actions, nor does it dismiss any instance of harm caused as a result of aversive procedures. On the contrary, the behavior-analytic field recognized limitations and acknowledged societal concerns, and new and improved practices were developed along with implementation guidelines (Axelrod, 1987). This pattern also has been observed in other professions. For instance, not too long ago, doctors prescribed arsenic and mercury to fight diseases. However, after discovering dangerous side effects associated with mercury, they began using less harmful drugs (e.g., penicillin); however, unlike mercury prescriptions, behavior reduction procedures are firmly based in science. Nevertheless, the behavior-analytic field assessed and identified negative side effects associated with punishment procedures, and the behavior-analytic ethical code now includes guidelines for punishment procedures (Behavior Analyst Certification Board [BACB], 2017).

The discontinuation of practices such as aversive punishment can partially be attributed to developing more effective, efficient, and safer function-based interventions (i.e., interventions addressing the reason an individual engages in the target behavior). The individualized approach to assessment and intervention helps practitioners

develop interventions matching the correct function rather than arbitrarily implement generic and potentially ineffective interventions. Function-based interventions also influenced the shift away from punishment-based interventions to reinforcement-based and proactive interventions within school settings. Historically, students in public school settings received reactive, aversive, and sometimes arbitrary consequences following target behavior

truly focused on individuality and person-centered plans, why make students comply with every demand? It seems that promoting compliance might inhibit personal development and independence. Does ABA focus on compliance or independence?

Behavior Analysis and Compliance

From a behavior-analytic perspective, a focus on compliance poses problems for two major reasons. First, the term

The individualized approach to assessment and intervention helps practitioners develop interventions matching the correct function rather than arbitrarily implement generic and potentially ineffective interventions.

(Sidman, 1989). In contrast, the shift away from aversive interventions involved teaching and rewarding appropriate behavior rather than administering arbitrary consequences following inappropriate behavior (Horner et al., 1990/2005). In fact, entire school districts implement programs based on positive reinforcement, such as positive behavior interventions and supports (PBIS).

Chantelle now recognized another similarity between education and behavior analysis; both fields included a history with punishment-based strategies, and both fields learned hard lessons from their histories. Furthermore, as each respective field grew and began to overlap, both educators and behavior analysts worked diligently to create, refine, and disseminate practices and guidelines intended to enhance the lives of people receiving behavior-analytic interventions. In her future practice, when possible, Chantelle knew she could avoid the use of aversive procedures (e.g., frequent reprimands, the removal of nonpreferred items or activities) by using function-based interventions to replace severe behavior with more socially acceptable behaviors. Chantelle wondered, though, if ABA

compliance may have different meanings across individuals and settings. For example, one teacher may interpret student compliance as generally following a set of rules within a given context, whereas another may view it as a particular response to a teacher's directive. Both interpretations of compliance reflect the performance of potentially different skills and may do little to identify multiple sources of noncompliance (e.g., language barriers, inability to perform the requested task). Consider a new teacher who reports a student to be "noncompliant" during group work activities. It is unclear as to what specific conditions the student is not responding (e.g., a request to begin a task or share materials, teacher's constructive feedback, classroom expectations to remain in close proximity to peers). Further, it is difficult to determine whether any lack of responding or "noncompliance" is a result of not understanding directions, a lack of social skills, or a history of escaping group work through avoiding participation. Overall, behavior intervention teams focus on clear descriptions of individuals' patterns of behavior that will help them more accurately determine the nature of a problem and subsequently address it.

Second, the criterion for demonstrating compliance is often set by the individual who directly benefits from compliant responding (e.g., the teacher arbitrarily determines whether or not a student's performance is acceptable). At best, this may produce inconsistent performance standards, thus making it difficult for students to understand expectations and ultimately be successful. At worst, it may present an ethical dilemma, as an individual's compliance may be more beneficial to others than to the individual engaging in compliant behavior (e.g., sitting quietly in one's desk instead of actually learning new skills). Further, in some cases, consequences for compliant behavior can put individuals at risk for harm and abuse (e.g., bullying behavior, inappropriate sexual contact). Thus, the focus of applications of ABA should be based upon a person-centered approach, and practitioners should remain keenly aware of the dangers associated with teaching individuals, especially those with disabilities, to freely comply with the requests of others.

Chantelle realized that using ABA didn't mean she needed to focus on compliance with demands. She could instead carefully consider goals and use ABA to help students exercise independence and decision making. She appreciated how behavior-analytic procedures were designed to provide individuals with ways to control their environment and lessen external influence on their choices. She was encouraged by all she had learned, but she had one final concern that was often raised by teachers at the conference—did ABA involve bribery? One teacher at the conference explained that a parent who observed his classroom became concerned that children were being bribed to follow simple rules. Each time a child answered a question or followed a direction, the consulting analyst gave the child a sticker.

Behavior Analysis and Bribery

Practitioners implementing behavior-analytic techniques often answer the

question of bribery after school consultation visits or classroom observations. For example, after a classroom observation, teachers or parents may ask, “Why did you bribe my child with a sticker for working when she should do it just because you asked her?” To which a teacher might reply, “Rewards, such as classroom cash, sticker charts, or point systems, teach students that hard work results in consequences, such as payment—much like for you and me.”

Individuals with and without disabilities engage in a range of behaviors as a result of consequences that often go unnoticed and would unlikely be considered bribery. For instance, Mr. Taylor’s paycheck produces consistent attendance at his teaching job, and Sandra’s positive interaction with her peers ensures her selection of a seat at her friends’ table during lunch. In both instances, it would be odd to suggest that the school’s principal or Sandra’s friends are using bribery to control behavior.

Despite perceived similarities between bribery and the use of positive reinforcement, there are stark differences between the two that largely stem from the systematic and programmed nature of positive reinforcement. Positive reinforcement is the introduction of a stimulus into an environment contingent on a specific behavior, resulting in the increased probability of that behavior occurring in the future after the introduction of the stimulus (Alberto & Troutman, 2013; Cooper et al., 2007). Bribery, on the other hand, is described by O’Leary, Poulos, and Devine (1972) as the promise of a reward before target behavior occurs in an attempt to influence the individual’s behavior.

In the previously presented example, the teacher noticed a student had difficulty responding to questions in class. The teacher then decided to give the student a sticker after she answered questions and noticed an immediate increase in her attempts to provide an answer. The teacher’s hope is then to systematically increase the

number of questions the student independently answers before receiving a sticker and, eventually, by pairing the stickers with praise and other natural consequences, that the student will begin answering the teacher’s questions for more typical reinforcers (e.g., positive feedback). However, bribery would occur if the teacher noticed the student off task and then began incentivizing the student by telling her, “Get back on task so that you can get your sticker,” rather than providing a sticker when the student was on task.

Although some educators may be uncomfortable with the programmed delivery of reinforcers, they must acknowledge that some students do not respond for the typical “pat on the back” or some “intrinsic motivator.” In these instances, teachers must consider adjusting their perceptions around “what students should do” and seek reinforcers that will serve to motivate their students. The goal of teaching is to create optimal conditions for learning. Personal preferences, prior experience, and level of competency all influence the way a person may engage within a learning environment. Further, many people work for a range of different reinforcers; successful teachers are able to identify what individual students need and then create instructional environments, which may include the programmed delivery of reinforcers, to support those learning needs.

However, behavior-analytic techniques also encourage using

functional and relevant responses that will access more natural reinforcers, such as the student’s developing friendships after learning appropriate ways to gain attention or learning math skills that are later used to purchase goods and services. Contrived reinforcers (e.g., programmed delivery of praise, stickers, or tokens) become a problem only when we overlay them on prevailing schedules of reinforcement for students who do not need that extra push.

Although appearing similar, Chantelle now distinguished between bribery and positive reinforcement. She also acknowledged the importance of understanding how reinforcement works and how to explain it in a way that will help others understand so they might also consider using reinforcement-based strategies.

Behavior-Analytic Methods and Guidelines for Use

After developing a greater understanding and clarifying behavior-analytic misconceptions, Chantelle inquired about specific behavior-analytic methods and how she could incorporate them into her future classroom. Although unable to cover the breadth of ABA’s utility in school settings, Dr. Thatcher was again encouraged by Chantelle’s curiosity and provided a brief overview of some behavior-analytic methods implemented in classrooms.

ABA does not dictate a “one-size-fits-all” approach to data collection but emphasizes behavior-analytic procedures that are doable and can be implemented by relevant stakeholders.

natural reinforcers (i.e., events likely to follow the student’s behavior in the natural environment; Hineline & Morris, 1992) and teaching students to wait for rewards rather than expect something right away. Ultimately, it is best practice to teach

Data Collection

Behavior intervention teams collect data using various procedures that are matched to a clearly defined target behavior of interest and the context under which the behavior is likely to occur. ABA does not dictate a

Table 1. Data Collection: Methods, Definitions, and Examples

Method	Definition	Example
Frequency	A count of how many times the behavior occurred	Recording how many times a student called out during math class
Percentage of occurrence	The number of correct responses divided by the number of opportunities to respond	Recording how many problems the student answered correct out of how many math problems total
Duration	How long the behavior occurs	Recording the minutes and seconds a student is out of the seat
Rate	Frequency of behavior in a specific amount of time	Recording how many times a student called out per minute during math class
Latency	The time between the presentation of a stimulus and the occurrence of the behavior	Recording the time between asking a student to begin a test and when the student begins the test
Interval recording	Estimates the frequency of behavior using one of three methods: time sampling, partial interval, or whole interval recording	Time sampling: Setting a timer for 10 minutes and recording whether the student was in seat when the interval ended Partial interval: Setting a timer for 5 minutes and recording whether the student left seat at any time during the interval Whole interval: Setting a timer for 2 minutes and recording whether the student remained in seat the entire 2 minutes

“one-size-fits-all” approach to data collection but emphasizes behavior-analytic procedures that are doable and can be implemented by relevant stakeholders. For example, it may not be practical for a general education teacher to constantly monitor a single student for an entire instructional period. A mindful behavior intervention team will collaborate to design a feasible data collection method that will capture meaningful data (see Table 1).

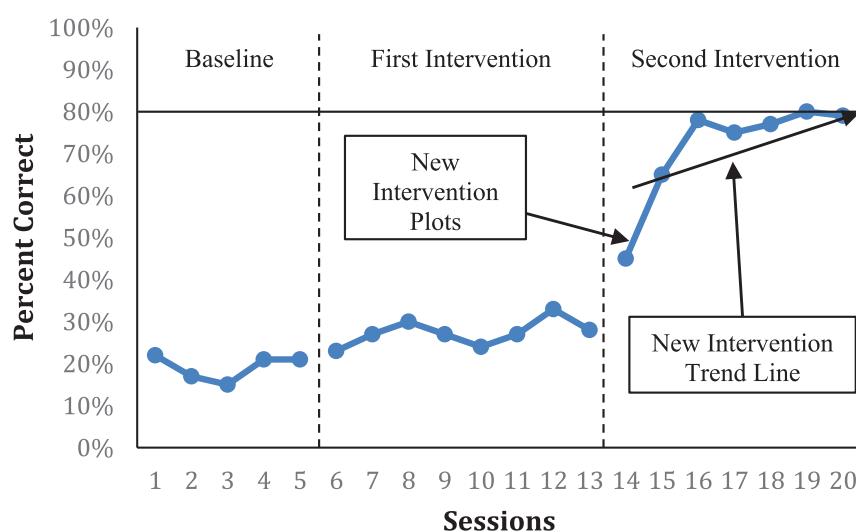
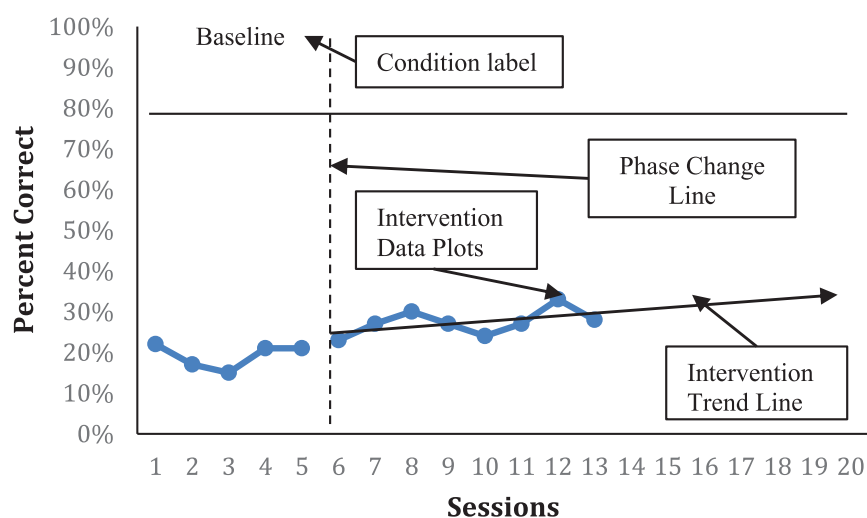
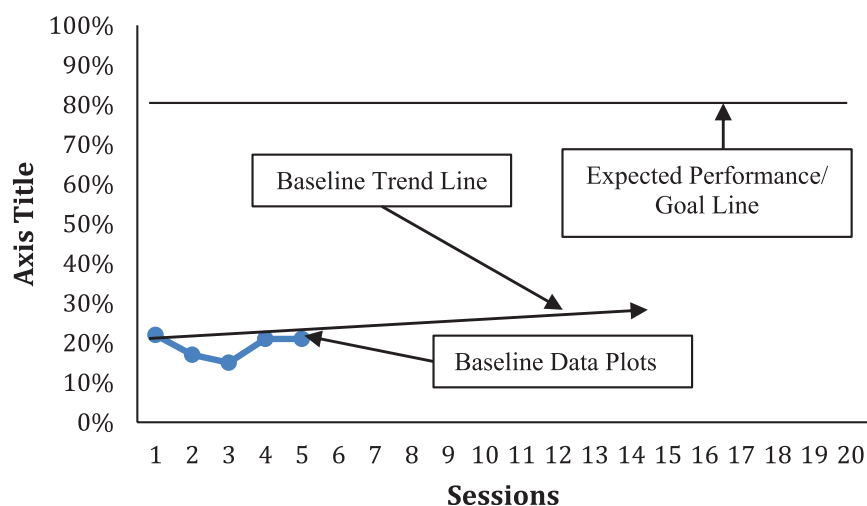
Most often, teachers concern themselves with how often a response occurs (i.e., frequency) and the percentage of time it occurs during instructional activities (i.e., percentage correct). For example, a teacher might want to know how many times a student with a communication disorder makes requests during the day or correctly names items within the environment. However, not all interventions focus on frequency of responding. Teachers may be interested in time spent in seat (i.e., duration), how many times per hour a student is out of seat (i.e., rate), or how quickly the student returns to the seat after a cue is given (i.e., latency). Collecting these data require ongoing

observation of the student, which may not always be feasible in large classrooms or in classrooms with multiple students receiving different educational programs. Therefore, data collection techniques that estimate behavior can be used. These collection techniques may include checking for the presence or absence of a behavior at the end of a short interval (i.e., time sampling) as well as recording whether or not a behavior occurs during a portion of time (i.e., interval recording). After appropriate data collection methods are identified, practitioners are expected to determine baseline levels of performance (i.e., the level of performance prior to intervention). This baseline performance represents the starting point against which practitioners will compare a student’s performance after beginning an intervention.

Teachers examine baseline data to determine when a student will achieve the desired level of performance. This is usually done by plotting the four or five baseline data points on a line graph and drawing a trend line that illustrates the direction of the baseline performance (see Figure 2). The teacher later plots

intervention data on the graph and draws a new trend line that illustrates current performance. If the objective were to increase a target behavior, the desired trend line would slope upward; however, if the goal were to decrease a target behavior, the desired trend line would slope downward. If intervention data indicate the skill will be mastered by the expected date, then the intervention is considered effective and implementation continues. If the plotted points indicate the intervention is ineffective or has negative effects (i.e., an undesired trend line), then it is altered or discontinued and a different intervention is adopted. The bottom panel in Figure 2 shows the progression of an intervention process in which baseline data are collected, an initial intervention was ineffective, and a second intervention was used that resulted in the expected performance. Phase change lines and condition labels indicate a condition change. For example, the first phase change line in the middle panel of Figure 2 indicates the beginning of an intervention, and the second phase change line in the bottom panel indicates the implementation of a different intervention.

Figure 2. Three panels of plotted data for percentage of time in seat for baseline condition, ineffective intervention condition, and effective intervention condition



Given the importance of data collection for the implementation of effective instruction, the following recommendations should guide the use of data collection:

1. Clearly define the target behavior.
2. Select an appropriate data collection method.
3. Establish the current level of responding by collecting baseline data.
4. Set a schedule for collecting and reviewing data once instruction or intervention begins.
5. Collect data and make changes to intervention as needed.

Punishment

As previously mentioned, the decreased use of aversive procedures can be attributed to developing effective function-based interventions; however, the shift away from aversive procedures also can be credited to BACB establishing, enforcing, monitoring, and frequently updating an ethical compliance code for behavior analysts (BACB, 2017). Although the ethical compliance code does not apply to non-Board Certified Behavior Analyst (BCBA) practitioners, it serves as a thorough document outlining BCBAs' conduct as well as their responsibility to their clients and the field. For instance, the document outlines a client's right to effective treatment, which includes using interventions with proven short- and long-term benefits. Behavior analysts also are required to "recommend reinforcement rather than punishment whenever possible" (BACB, 2017, p. 13), and when punishment procedures are required, behavior analysts must include a reinforcement component for an alternative behavior.

Teachers also can follow the BACB's guidance by asking the following questions prior to considering the use of punitive or more restrictive

procedures: (a) whether there is a behavior that can be taught or strengthened that will functionally replace the target behavior, and (b) whether the environment can be adjusted so that the target behavior is no longer necessary. For example, when attempting to decrease the number of times a student calls out during class, a teacher might teach the student to raise a hand as an alternative method to obtain the teacher's attention and ignore talk-outs instead of delivering derisive statements. Ultimately, interventions are designed to help students meet their needs by teaching and reinforcing functional skills.

Function-Based Assessments

Prior to designing behavior reduction interventions, an educational team should conduct a functional behavioral assessment (FBA). An FBA involves data collection methods aimed at evaluating environmental variables and their relation to the target behavior. These assessment methods use a person-centered approach and do not treat behaviors such as aggression (e.g., hitting, kicking, spitting) the same way for every student who engages in aggressive behaviors. Instead, practitioners determine why the person engages in aggression toward others and create function-based interventions (i.e., interventions addressing why the student engages in the target behavior) to help students express their needs in more socially appropriate ways (Cooper et al., 2007). For example, after thorough assessment, an intervention team might determine Walden hits his teacher when he is trying to escape a demand, but Eleanor hits her mother when she wants to play on the tablet. Instead of implementing the same procedures, or just a punishment-based program to decrease hitting, the team might help Walden learn to ask for a break and teach Eleanor to ask for a turn on the tablet.

Behavior analysts, teachers, paraprofessionals, students, teachers, and other stakeholders should all be key players of a collaborative team

when conducting FBAs. Furthermore, it is important to not only understand the relation between the target behavior and environmental variables but to also understand the target behaviors' impact on those involved in these

leading to a hypothesis about the function of the target behavior (i.e., why the target behavior occurs). For example, direct observations might indicate that the target behavior occurs in a specific context or during a particular time of

Furthermore, it is important to not only understand the relation between the target behavior and environmental variables but to also understand the target behaviors' impact on those involved in these experiences, including the individual.

experiences, including the individual. Therefore, FBAs often involve indirect measures, such as parent or guardian questionnaires. The questionnaires primarily focus on events occurring immediately before and after the target behavior occurs, but they also often involve evaluating additional factors that might not be as obvious, such as sleep deprivation, medication changes, or changes in family structure (e.g., the addition of a newborn). Interview questions might include the following:

1. Where is the target behavior most and least likely to occur?
2. With whom is the target behavior most and least likely to occur?
3. What generally happens before, during, and after the target behavior occurs?
4. How does the target behavior impact the individual's, and other family members', daily routines? For example, is the individual unable to attend certain events because of the target behavior?
5. Describe the individual's morning, afternoon, and evening routines.

In addition to indirect assessments, such as questionnaires, an FBA also requires the behavior intervention team to conduct direct observations. For example, ABC (antecedent, behavior, consequence) data forms provide a format for practitioners to record events and other relevant information occurring before, during, and after the behavior occurs (Cooper et al., 2007). Eventually, this information might reveal a pattern

day, or more explicit pattern details might emerge, such as target behavior occurring after the individual is denied access to a preferred item. Although antecedents (e.g., events occurring before target behavior) are important to note, the consequence (i.e., events occurring after the target behavior occurs) influences whether or not the behavior will continue to occur (Skinner, 1938). For example, after all teachers entered grades earlier than expected, a principal allowed teachers to wear jeans. In this scenario, if wearing jeans is a preferred activity for the teachers, the jeans-day provision will likely influence early grade entry in the future.

Furthermore, events occurring after the target behavior occurs can be classified into a few categories (e.g., access to an item or attention, escape from work or an undesired social situation, or automatic stimulation; Cooper et al., 2007). For example, after conducting direct observations, a data collector might note that during group instruction, Emily called out to answer a question, rather than raising her hand, eight out of nine times. This information, combined with indirect measures, might lead to a hypothesis indicating access to peer attention as a potential reason Emily engages in yelling behavior. Identifying peer attention as a potential reinforcer will assist the behavior intervention team when creating an effective function-based intervention. Without identifying peer attention as the reason behind Emily's yelling, the teacher might attempt ignoring

Table 2. Group Contingencies

Type	Example
Dependent: A group contingency in which the entire group earns a reward based on the behavior of one or more group members (e.g., Jones, Boon, Fore, & Bender, 2008).	1. Prior to the lesson, the teacher a. Reviews and practices classroom rules. b. Secretly selects a student. c. Determines number of observation checks. d. Determines an obtainable point goal (e.g., 6 is not an obtainable point goal if the teacher plans to check the student's behavior five times). 2. During the lesson, the teacher a. Periodically observes the student during intervals. b. Observes whether the student is following predetermined rules. c. Provides a point during observation checks if the student is following all rules but does not provide a point if the student is not following all rules. 3. After the lesson, a. If meeting the goal, the teacher reveals the student and provides a reward to the entire class. b. If the student does not meet the goal, the teacher does not reveal the student and states the mystery student did not meet the goal.
Interdependent: A group contingency in which the entire group earns a reward if every group member follows the rules (e.g., Kamps et al., 2011).	1. The teacher follows the same steps as the dependent group contingency, except a. The teacher announces the groups at the beginning of the game. b. During planned checks, the teacher observes and records whether or not an entire group met the criteria rather than an individual. c. Groups receive rewards based on the group's performance rather than an individual's.
Independent: A group contingency in which all individuals meeting the criteria receive a reward (e.g., Chazin & Ledford, 2016)	1. The teacher tells the class, "Every student earning an A on the test will receive an extra homework pass." 2. The teacher provides the homework pass to all students who scored an A on the test.

Emily's yelling, which would likely have little to no effect on Emily's yelling due to the value of peer attention. However, after identifying the appropriate function, the behavior intervention team can develop interventions to reduce peer attention, such as a group behavior game rewarding students for ignoring inappropriate behavior (e.g., classwide function-related intervention teams; Kamps et al., 2011). In Table 2, additional group contingency examples are provided.

Functional Communication Training

Practitioners (teachers, interventionists) implementing behavior-analytic techniques are most interested in increasing individuals' independence or freedom to choose,

negotiate for, and access preferred items and activities within their environment and limiting their exposure to aversive situations. This behavior-analytic, person-centered approach to intervention may be best exemplified in the application of functional communication training (FCT). Carr and Durand (1985) developed FCT to reduce challenging behavior while simultaneously honoring an individual's need to communicate. Instead of merely trying to reduce a target behavior, FCT asks practitioners to determine why the individual is engaging in the behavior and to then provide the individual with a conventional and likely more effective way to ask for assistance. FCT has been established as an evidence-based practice for use with individuals with developmental disabilities (Wong et al., 2015).

In general, FCT has five steps that must be followed to ensure success (Table 3). First, the practitioner conducts an FBA to identify current reinforcers maintaining challenging behavior (e.g., access to the attention, escape from tasks). This FBA process is generally composed of interviews, direct observation, and in some cases, the manipulation of environmental variables (e.g., how a teacher responds to a student's behavior). Second, the communicative response that serves the same function as the challenging behavior is identified. For example, if an FBA indicates a student's challenging behavior is to escape difficult tasks, then a teacher might have the student touch a card with the words "I want a break" as the alternate functional communicative response (FCR). It is important that this response require minimal effort

Table 3. Functional Communication Training Implementation

Step	Description
1. Identify reinforcers that maintain challenging behavior	• Use indirect assessment ◦ Teacher-guardian questionnaires • Use direct observations ◦ ABC data • Identify potential reinforcers ◦ Escape from work or uncomfortable social situation ◦ Access to an activity or item
2. Identify a functional communicative response (FCR)	• AAC device, gesture, picture exchange, or vocalization indicating a desired break, attention, or item • A code system, such as a red-and-green card on the student's desk ◦ Red indicates a break request ◦ Green indicates attention or help request • Raising one's hand instead of calling out
3. Teach the FCR	• Teach in natural settings and when the motivation is present • Initially, the student might require prompting to acquire the FCR • Contrive situations to allow the student more opportunities to practice ◦ For example, if the student escapes work by tearing up worksheets, place the work on the student's desk and prompt the student to request a break
4. Apply differential reinforcement	• Reinforce FCR and weaken response to problem behavior ◦ Attention-maintained example: If the student raises a hand instead of blurting out, the teacher praises the student and provides high-quality attention, but if the student blurts out, the teacher provides limited or low-quality attention ◦ Escape-maintained example: If the student tears up spelling work to escape math, the teacher continues to provide math work and prompts the student through completion, but if the student requests a break, the teacher removes the work and allows a break
5. Gradually fade reinforcement for the FCR	• Monitor data to determine if, when, and how to begin fading reinforcement, but remember not all reinforcement schedules should be faded (e.g., a teacher's paycheck is not faded) ◦ Rather than allowing a break every time the student requests a break, allow a break every other time or set a work requirement before a break is available ◦ Rather than calling on students every time they raise their hand, the teacher calls on them every other time, then every third time, and so on • Continuously monitor data to determine changes in the reinforcement schedule (e.g., if the schedule was faded too quickly)

Note. ABC = antecedent, behavior, consequence; AAC = augmentative and alternative communication.

and be understood by all available listeners. The goal is to make the FCR more effective than the challenging behavior in helping the student obtain a desired outcome. Third, the teacher must take time to teach the FCR and make sure that it rests firmly in the student's repertoire. This often will require directly teaching the response within environments similar to those in which the target behavior occurs. For example, when teaching a student to ask for a break, the teacher might present a difficult task and, when observing precursors to challenging

behavior (e.g., stops working, talking under one's breath), prompt the student to ask for a break. Fourth, the teacher must then apply differential reinforcement, that is, respond immediately and consistently to the FCR and lessen the response or not respond at all to challenging behavior (Tiger, Hanley, & Bruzek, 2008). For example, if the student requests a break, the work requirement is immediately removed, but the work requirement remains in place if the student engages in the target behavior. Finally, as it is not practical to provide

continuous access to reinforcers, the teacher must systematically fade reinforcement by employing one of several fading techniques. For instance, after the student masters asking for a break rather than engaging in the target behavior, the teacher might introduce a short waiting period.

Antecedent Strategies

To enhance the effects of positive reinforcement, teachers might use antecedent strategies (i.e., strategies

Table 4. Antecedent Strategies

Strategy	Example	Rationale
Choices	Provide the student with location, materials, person, or activity choices • Work at the table or desk? (location) • Write with the pencil or pen? (materials) • Read with Ms. Ti or Mr. Pat? (person) • Write words or do math? (activity)	This strategy respects the learner's autonomy while simultaneously providing the learner's right to quality instruction.
High-probability request sequence	Ask a number of easy demands the student usually completes and then follow with a harder demand • Get the blue pencil. • Draw a circle. • Underline the letter <i>w</i>. • Copy the word <i>water</i>. (target skill)	The learner experiences success while completing known skills, which decreases the likelihood the learner will experience behavior consistent with frustration.
Embed demands into preferred activities	While a student is engaged in a preferred activity, make academic requests. Lesson targets: prepositions, opposites, actions, receptive language, color identification, pronouns • Put the car under [on top, beside]. • Show me how the car drives fast [slow]. • What color is my [your, his, her] car?	This strategy incorporates functional or academic skills into a preferred activity. As a result, skill acquisition is paired with fun and enjoyment.

in place to reduce the likelihood of the target behavior occurring), which have the potential to decrease an individual's target behaviors while simultaneously increasing their use of contextually-appropriate behaviors. Examples of antecedent strategies include the provision of choices, high-probability request sequences, and embedding demands into preferred activities (Miltenberger, 2006; Table 4).

One specific example of an antecedent strategy is a "first-then" statement (see Table 5). A first-then statement specifies the contingency before a demand or task is placed and informs the student what needs to be done before the student can access reinforcement. For example, when the student is working independently on a math problem, a teacher might state, "First, do your math problems; then, you can use the computer." Similar models are used in everyday life for adults—when the school district provided Mr. Taylor with a contract, the district essentially made a first-then statement: "First, work; then, you get a paycheck."

At the end of their meeting, Dr. Thatcher smiled in response to his student's curiosity and said, "In your future special education practice, ABA can be used for teaching academic, social, and adaptive skills across different populations and settings. It can even be used to improve your own practice, to improve the practice of your paraprofessionals, and to make your entire school a more effective and welcoming place for students. If you decide to become a behavior-analytic practitioner, stay curious and continue to seek answers to your questions."

As Chantelle walked away from the meeting, she reflected on the many ABA practices she had observed in her practicum placements. Although she understood and in some cases empathized with the conference speakers' concerns related to the misapplication of ABA principles, she remembered that ABA is the science of predicting and changing behaviors important to those involved, including the individual. She also felt that the benefits of adopting an ABA approach in her future classroom certainly outweighed any risks. She felt inspired

to develop these new, powerful, person-centered tools to be the best teacher she can be.

Conclusion

ABA, or the application of behavioral principles to important social issues, has long served a crucial role in programming for special education students. Its procedures have been well established in the research literature and many deemed as evidence-based practices for students with and without disabilities. Unfortunately, it is often subject to misconceptions (e.g., it is only for students with ASD) and in some cases associated with a past marred by the use of unsavory practices (e.g., restraint, electric shock). However, the adoption of behavior-analytic practice can be an important tool for teachers. ABA complements any educational endeavor in that it places the learner front and center within intervention and instructional design and demands that educators are accountable and resilient in supporting students in achieving the highest quality of life possible.

Table 5. First-Then Statements: Tips, Rationale, Examples, and Nonexamples

Tip	Rationale	Example	Nonexample
1. Identify a reinforcer.	The statement will be effective only if the reinforcer (the “then” part of the statement) is something the student wants.	Asking the student what he or she would like to work for before making the first-then statement. “Do you want to work for stickers or coloring?”	Selecting an item for the “then” statement because it is close or convenient. Stating “... then pretzels” because you have pretzels in your bag.
2. Use simple, precise language.	Directions should be clear, be simple, and use as few words as possible to avoid confusion.	“First, math work; then, break.”	“First, you need to sit down and do your math problems for a little while; then, you can go over to the play area and have a break.”
3. Increase the salience of the contingency.	The contingency (what the student has to do to earn a reward) should be clear. Adaptations, such as corresponding pictures, may be incorporated to increase the salience for students needing more support.	Incorporating pictures to increase saliency. For the statement, “First, work; then, Skittle,” the teacher may display corresponding pictures (picture of the student’s work box, picture of Skittles).	Repeating the statement, adding words. “First, work; then, Skittle. You need to do your work before you get the Skittle. No, work first; then, Skittle. No Skittle if you don’t work.”
4. Prepare the environment.	Arrange the environment so the reward can be delivered immediately after the demand (“first” part of the statement) is completed.	Ms. White wears a fanny pack during the school day so she can keep edible reinforcers stocked and nearby throughout the day.	Ms. Greenway made the statement, “First, write letters; then, pretzels.” After Anna wrote her letters, Ms. Greenway realized she left the pretzels in the car.
5. Be flexible.	Students may change their mind about what they want after complying with the demand. If they no longer want the originally stated reward, that is okay. Allow them to select something different.	Emily asked to work for M&M’s, and her teacher told her, “First, calendar; then, M&M’s.” After calendar, Emily saw that Tiffany earned Doritos and asked for Doritos instead. Her teacher let her trade M&M’s for Doritos.	Mr. Taylor told Lauren, “First, wipe tables; then, stickers.” After Lauren wiped the tables, she asked Mr. Taylor if she could color. Mr. Taylor said, “No, you asked for stickers.”

References

- Alberto, P. A., & Troutman, A. C. (2013). *Applied behavior analysis for teachers* (9th ed.). Columbus, OH: Pearson.
- Axelrod, S. (1987). Functional and structural analyses of behavior: Approaches leading to reduced use of punishment procedures? *Research in Development Disabilities*, 8, 165–178. doi:10.1016/0891-4222(87)90001-1
- Bailey, J., & Burch, M. (2011). *Ethics for behavior analysts* (2nd ed.). New York, NY: Routledge.
- Baller, J. B., Barry, C. L., Shea, K., Walker, M. M., Ouellette, R., & Mandell, D. S. (2016). Assessing early implementation of state autism insurance mandates. *Autism*, 20, 796–807. doi:10.1177/1362361315605972
- Behavior Analyst Certification Board. (2017). *Professional and ethical compliance code for behavior analysts*. Retrieved from <https://www.bacb.com/ethics/ethics-code/>
- Cagliani, R. R., Ayres, K. M., Whiteside, E., & Ringdahl, J. E. (2017). Picture Exchange Communication System and delay to reinforcement. *Journal of Developmental and Physical Disabilities*. Advance online publication. doi:10.1007/s10882-017-9564-y
- Carr, E. G., & Durand, V. M. (1985). Reducing behavior problems through functional communication training. *Journal of Applied Behavior Analysis*, 18, 111–126. doi:10.1901/jaba.1985.18-111
- Carroll, R. A., Joachim, B. T., St. Peter, C. C., & Robinson, N. (2015). A comparison of error-correction procedures on skill acquisition during discrete-trial instruction. *Journal of Applied Behavior Analysis*, 48, 257–273. doi:10.1002/jaba.205
- Chazin, K. T., & Ledford, J. R. (2016). Independent group contingencies. In *Evidence-based instructional practices for young children with autism and other disabilities*. Retrieved from <http://vkcc.mc.vanderbilt.edu/ebip/independent-group-contingencies>
- Collins, B. C., Terrell, M., & Test, D. W. (2017). Using a simultaneous prompting procedure to embed core content when teaching a potential employment skill. *Career Development and Transition for Exceptional Individuals*, 40, 36–44. doi:10.1177/2165143416680347
- Cooper, J. O., Heron, T. E., & Heward, W. L. (2007). *Applied behavior analysis* (2nd ed.). Upper Saddle River, NJ: Pearson.

- Council for Exceptional Children. (2015). *What every special educator must know: Professional ethics and standards* (7th ed.). Arlington, VA: Author.
- Fixsen, D. L., Phillips, E. L., & Wolf, M. M. (1972). Achievement place: The reliability of self-reporting and peer-reporting and their effects on behavior. *Journal of Applied Behavior Analysis*, 5, 19–30. doi:10.1901/jaba.1972.5-19
- Guercio, J. M., & Murray, W. J. (2014). Licensure for behavior analysts: The path to responsible and cooperative action. *Behavior Interventions*, 29, 225–240. doi:10.1002/bin.1388
- Horner, R. H., Dunlap, G., Koegel, R. L., Carr, E. G., Sailor, W., Anderson, J., ... O'Neill, R. E. (2005). Toward a technology of “non-aversive” behavioral support. *Research & Practice for Persons with Severe Disabilities*, 30, 3–10. (Reprinted from *Journal of the Association for the Severely Handicapped*, 15, 125–132, 1990) doi:10.1177/154079699001500301
- Hineline, P. N., & Morris, E. K. (1992). Natural reinforcement: A way to improve education. *Journal of Applied Behavior Analysis*, 25, 71–75.
- Jones, M., Boon, R. T., Fore, C. I., & Bender, W. N. (2008). “Our mystery hero!” A group contingency intervention for reducing verbally disrespectful behaviors. *Learning Disabilities: A Multidisciplinary Journal*, 15(2), 61–69.
- Kamps, D., Wills, H. P., Heitzman-Powell, L., Laylin, J., Szoke, C., Petrillo, T., & Culey, A. (2011). Class-wide function-related intervention teams: Effects of group contingency programs in urban classrooms. *Journal of Positive Behavior Interventions*, 13, 154–167. doi:10.1177/1098300711398935
- Lovaas, O. I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. *Journal of Consulting and Clinical Psychology*, 55, 3–9. doi:10.1037/0022-006X.55.1.3
- Miltenberger, R. (2006). Antecedent interventions for challenging behaviors maintained by escape from instructional activities. In J. Luiselli (Ed.), *Antecedent assessment and intervention* (2nd ed., pp. 101–124). Baltimore, MD: Brookes.
- O’Leary, K. D., Poulos, R. W., & Devine, V. T. (1972). Tangible reinforcers: Bonuses or bribes? *Journal of Consulting and Clinical Psychology*, 38, 1–8. doi:10.1037/h0032412
- Roane, H. S., Ringdahl, J. E., & Falcomata, T. S. (2015). *Clinical and organizational applications of applied behavior analysis*. San Diego, CA: Elsevier Academic Press.
- Sidman, M. (1989). *Coercion and its fallout*. Boston, MA: Authors Cooperative.
- Skinner, B. F. (1938). *The behavior of organisms: An experimental analysis*. Oxford, UK: Appleton-Century.
- Tiger, J., Hanley, G., & Bruzek, J. (2008). Functional communication training: A review and practical guide. *Behavior Analysis in Practice*, 1, 16–23. doi:10.1007/BF03391716
- Van Houten, R., Axelrod, S., Bailey, J. S., & Favell, J. E. (1988). The right to effective behavioral treatment. *Behavior Analyst*, 11, 111–114. doi:10.1901/jaba.1988.21-381
- Wong, C., Odom, S., Hume, K., Cox, A., Fettig, A., Kucharczyk, S., ... Schultz, T. (2015). Evidence-based practices for children, youth, and young adults with autism spectrum disorder: A comprehensive review. *Journal of Autism and Developmental Disorders*, 45, 1951–1966. doi:10.1007/s10803-014-2351-z

Cary E. Trump, M.Ed., BCBA, Doctoral Student, Communication Sciences and Special Education, University of Georgia, **Robert C. Pennington, PhD** BCBA-D, Associate Professor, Department of Special Education, University of North Carolina-Charlotte, **Jason C. Travers, PhD**, BCBA-D, Associate Professor, Department of Special Education, University of Kansas, **Joel E. Ringdahl, PhD**, BCBA, Associate Professor, Communication Sciences and Special Education, University of Georgia, **Erinn E. Whiteside, M.A.T.**, BCBA, Doctoral Student, Communication Sciences and Special Education, University of Georgia, and **Kevin M. Ayres, PhD**, BCBA-D, Professor, Communication Sciences and Special Education, University of Georgia.

Address correspondence concerning this article to Cary E. Trump, Communication Sciences and Special Education, University of Georgia, 514 Aderhold Hall, 110 Carlton St., Athens, GA 30602 (e-mail: carytrump@gmail.com).

TEACHING Exceptional Children, Vol. 50, No. 6, pp. 381–393. Copyright 2018 The Author(s).

Copyright of Teaching Exceptional Children is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.